A Unified Data-Aware Fidelity and Regularization Learning Paradigm for Thick Cloud Removal of Multitemporal Remote Sensing Images

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***Abstract*—Thick cloud removal is a long-standing and critical challenge in remote sensing (RS) image processing, with the increasing availability of multitemporal RS images offering new opportunities to address this problem. The main limitation of existing cloud removal methods is that the classical fidelity only considers original pixel domain or handcrafted/pretrained filtered domains, overlooking the individuality filters and the corresponding feature behind each RS image, which leads to evident detail discrepancies. To address this issue, we suggest a data-aware fidelity based on the untrained neural network, which encourages deep data-aware feature matching between the contaminated image and the guidance image. Complementary to the data-aware fidelity, we design the deep self-representation to implicitly impose regularization, benefiting from the same untrained neural network. Equipped with the elaborately designed fidelity and regularization, we propose a unified data-aware fidelity and regularization learning (DAFRL) paradigm for thick cloud removal that flexibly adapts to diverse multitemporal RS images. Under this paradigm, the fidelity and regularization are empowered by the same untrained neural network, serving distinct functions while collaborating organically. Experimental results on both simulated and real datasets show that the proposed DAFRL effectively preserves fine details and outperforms the compared methods.**

***Index Terms*—Data-aware fidelity, deep self-representation, multitemporal remote sensing (RS) images, thick cloud removal, untrained neural network.**

I. Introduction

With the rapid development of Earth observation technology, remote sensing (RS) images are widely used as the core of technical analysis in a variety of fields, including environmental monitoring [1], [2], land cover analysis [3], [4], and so on. However, RS images are often disturbed by thick clouds when captured by optical satellites, leading to the loss of information and thus affecting their application in downstream missions, such as segmentation [5], unmixing [6], and classification [7], [8]. Therefore, it is crucial to develop an effective thick cloud removal method to reconstruct high-quality RS images to enhance subsequent applications [9].

In traditional cloud removal techniques [10], the spatial-based, spectral-based, and temporal-based methods are widely adopted due to their simplicity and computational efficiency. These methods primarily rely on a single source of complementary information to reconstruct the missing or contaminated areas in RS images. Spatial-based methods leverage spatial correlations within a single image, assuming that the structure and texture of cloud-contaminated areas can be inferred from adjacent cloud-free areas [11], [12]. Spectral-based methods exploit the consistency of spectral signatures across different bands, utilizing spectral relationships to predict the missing information [13], [14], [15]. Temporal-based methods utilize historical or multitemporal observations to reconstruct cloud-contaminated areas by referencing information from cloud-free areas taken at different time nodes [16], [17]. Although these methods can yield promising results, their reliance on a single source of complementary information limits their applicability, often constraining them to specific tasks under restricted conditions.

To compensate for these shortcomings, the hybrid-based methods achieve satisfactory reconstructed performance by fusing two or more of the spatial, spectral, and temporal information. By leveraging the complementary advantages of these different modalities, hybrid-based methods enhance the reconstruction quality and improve the robustness of cloud removal techniques in RS applications [18]. Most of these methods focus on improving the regularization term, such as low-rank modeling [19], [20], smoothness property [21], and deep image prior [22]. More recently, the pioneer works design fidelity under transform domains (e.g., gradient domain [23] and pretrained deep domain [24]) to reconstruct multitemporal RS images with promising quality. However, existing fidelity terms under fixed transform domains may fail to adaptively capture the unique features inherent in RS images, limiting personalized detail reconstruction results. Consequently, the underlying challenge is how to develop a thick cloud removal

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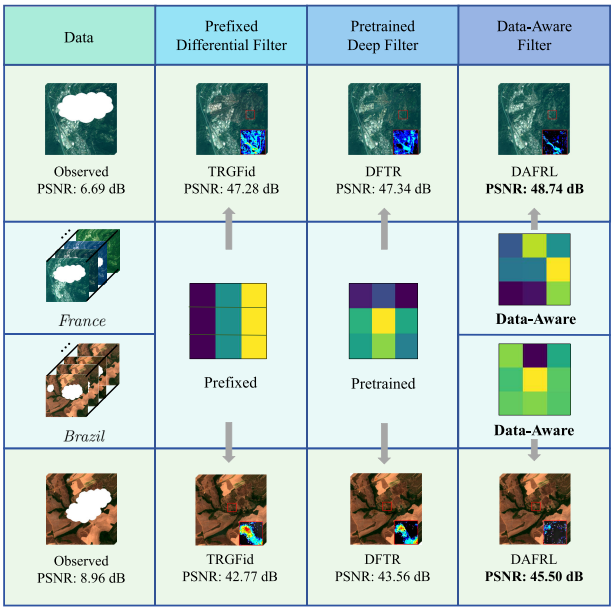


Fig. 1. Comparison of cloud removal results based on different filters. For different data, the prefixed differential filters (e.g., TRGFid) and pretrained deep filters (e.g., DFTR) are fixed, while the proposed data-aware filters (DAFRL) are more flexible. The residuals corresponding to the regions marked by red rectangles in each reconstructed image are enlarged to enhance the visual contrast effect.

framework that better exploits the specificity of multitemporal RS images to reconstruct fine details.

To meet this challenge, we propose a unified learning paradigm for thick cloud removal in multitemporal RS images, which simultaneously accomplishes data-aware fidelity and exploits the implicit regularization from deep self-representation within the same untrained network. Concretely, we employ an end-to-end untrained neural network to extract the unique features of multitemporal RS images. Inspired by the powerful feature extraction capabilities of deep neural networks [25], we encourage having similar feature representations between reconstructed images and guidance images in the data-aware deep domain, instead of the prefixed domain or pretrained deep domain, using the data-aware fidelity. Compared to prefixed and pretrained filters, the proposed data-aware filters can be more flexible in focusing on the unique features of each data; see Fig. 1. Meanwhile, the self-representation models the relationships among data samples by expressing each point as a linear combination of others within the same subspace, making it an appropriate candidate for regularization in reconstruction tasks. Coupled with the ability of deep networks to capture significant amounts of image statistics, we design a deep self-representation based on the same untrained neural network to automatically implement an implicit regularization, replacing the handcrafted regularization terms with improved results. By jointly embedding both fidelity and regularization constraints within a unified neural network, the learning paradigm ensures a synergistic interplay between feature alignment and structural coherence, which enjoys advantageous performance in terms of both robustness and accuracy.

To summarize, this work makes the following contributions.

1) To leverage the unique characteristics of diverse multitemporal RS images, we suggest a data-aware fidelity to extract deep data-aware features of multitemporal RS images, while introducing the implicit regularization derived from deep self-representation based on the same untrained neural network, leading to more accurate reconstruction results of multitemporal RS images.
2) By integrating the well-designed fidelity and regularization, we propose a novel thick cloud removal framework for the first time, unifying data-aware fidelity term and implicit regularization term in an unsupervised learning paradigm [termed as data-aware fidelity and regularization learning (DAFRL)]. Experimental results on multiple datasets demonstrate that the proposed cloud removal method outperforms several existing methods.

The remainder of this article is organized as follows. Section II reviews the related work. In Section III, we introduce our DAFRL model and solving algorithm. Section IV provides the experimental results, followed by discussions. The conclusion is given in Section V.

II. Related Works

A. Spatial-Based Methods

The spatial-based methods mainly use the spatial correlation between cloud-free regions and cloud-contaminated regions in a single RS image to reconstruct missing areas, which shares conceptual similarities with the image inpainting problem [26]. A simple and intuitive strategy for thick cloud removal is to employ interpolation methods, such as kriging interpolation [11] and nearest neighbor interpolation [12]. In addition, the exemplar-based methods [27], [28], the maximum a posteriori-based method [29], and the bandelet-based method [30] exploit the interrelationships between pixels in the spatial dimension to achieve considerable reconstruction results. Recently, deep learning (DL) has evolved rapidly and achieved impressive results in the field of RS image processing. VPint2 [31] is a data-driven spatial interpolation method that removes clouds without prior training by propagating information. Unlike conventional image reconstruction methods, which directly use pixels surrounding the contaminated region to fill in missing areas, deep convolutional neural networks (CNNs) combine neighboring pixels and global information through convolutional computations. For example, Zheng et al. [32] employed a generative adversarial network to maintain visual consistency for the single-image cloud removal task. However, while spatial-based methods are typically effective for addressing small-scale cloud contamination, their performance is limited in reconstructing large areas of cloud contamination due to insufficient reference information.

B. Spectral-Based Methods

The spectral-based methods provide an effective solution for cloud removal tasks by leveraging the spectral correlation inherent in RS images. For instance, the moderate resolution imaging spectroradiometer (MODIS) bands 6 and 7 exhibit strong correlations, which allow for the estimation of missing data through polynomial linear fitting [13]. Building on this principle, advanced methods, such as local least-squares fitting combined with histogram matching [14] and within-class local fitting [15], have been proposed to further

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enhance reconstruction accuracy by considering scene-specific relationships between spectral bands. Furthermore, Eckardt et al. [33] introduced a closest feature vector technique to remove clouds based on the closest spectral fit [34]. Li et al. [35] utilized the information from all the bands to reconstruct the image through the robust multilinear regression. These methods exploit the relationships between spectral bands to reconstruct missing data, offering a promising approach for cloud removal. However, single-temporal RS images often face the challenge of simultaneous cloud cover affecting all spectral bands, resulting in a complete loss of information in the corresponding region, which limits the applicability of these methods in such cases.

C. Temporal-Based Methods

The temporal-based methods focus on exploring the correlation of multitemporal RS images acquired at different time nodes to effectively remove cloud contamination [16]. In the literature, one common approach [17] in temporal-based cloud removal is to replace the degraded pixels with estimations derived from a regression model that utilizes corresponding pixels from RS images captured at other time nodes. Beck et al. [36] estimated vegetation dynamics using a double logistic function applied to a time series of RS images. Li et al. [37] presented a nonnegative matrix factorization technique to remove cloud cover from contaminated images by leveraging multitemporal data. Besides, the machine learning [38] and dictionary learning [39] have been increasingly applied to the cloud removal task to exploit the relationship between the original and temporal data, improving the reconstruction accuracy. Wang et al. [40] employed convolutional LSTM and residual decoding to reconstruct cloudy time-series images. Nonetheless, the temporal-based methods often fail to fully exploit the structural and contextual information within each individual image, which is essential for accurate reconstruction in severely degraded regions. Therefore, relying solely on temporal information to remove the cloud remains insufficient.

D. Hybrid-Based Methods

The hybrid-based methods integrate spatial, spectral, and temporal information of multitemporal RS images to enhance the cloud removal process, overcoming the limitations of relying on a single source of information [41]. To fully utilize the multisource of complementary information, a wider range of methods is employed. For example, Zhang et al. [42] introduced a tensor nuclear norm (TNN)-based method combined with a deep prior to remove the thick clouds. Xu et al. [43] proposed a stacked nonlinear Tucker decomposition (STD) to capture complex structures in an RS image. He et al. [21] imposed the tensor ring decomposition model with a total variation (TV) regularizer on multitemporal RS image recovery. Zheng et al. [44] removed the thick cloud of multitemporal RS images with a bilateral tensor ring. All of the above work advanced the design of regularization terms to improve reconstruction, while neglecting the design of the fidelity term.

Recently, Li et al. [23] presented the gradient domain fidelity to carve textures of multitemporal RS images. Beyond that, the pretrained deep domain fidelity [24] further extracted essential features through the deep neural network. However, these fixed transform domain fidelity terms are still insufficient to capture the unique characteristics of RS images, limiting the results of reconstruction. Comparatively, our method provides a data-aware fidelity with an implicit regularization to extract dynamic features via the unified learning paradigm, which is more efficient and flexible for multitemporal RS images reconstruction.

III. Proposed Method

Traditional cloud removal methods have primarily focused on refining regularization terms to enhance reconstruction accuracy while often overlooking the importance of designing the fidelity term. Existing pioneering cloud removal methods rely on a fixed formulation of the fidelity term, which fails to account for the unique structural and characteristics of individual RS images. This limitation leads to significant detail discrepancies, as the fixed transform domain fidelity term fails to capture image-specific variations, resulting in suboptimal reconstruction of fine details. To tackle these problems, we propose a DAFRL paradigm to reconstruct high-quality RS images using a shared untrained neural network by fusing spatiotemporal information from multitemporal RS images. The workflow of the proposed DAFRL is provided in Fig. 2.

A. Notations and Preliminaries

In this subsection, we summarize some basic notations used in our article. Scalars, vectors, matrices, and tensors are denoted by x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$, respectively. The element of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ in position $(i_1, \ldots, i_N)$ is denoted by $\mathcal{X}(i_1, \ldots, i_N)$, and similarly for vectors and matrices, i.e., $\mathbf{x}(i)$ and $\mathbf{X}(i_1, i_2)$. The Frobenius norm of Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is denoted by $\|\mathcal{X}\|_F = \left(\sum_{i_1, \ldots, i_N} |\mathcal{X}(i_1, \ldots, i_N)|^2\right)^{1/2}$. The multitemporal RS image can be expressed as the fourth-order tensor $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$, where H is spatial height, W is spatial width, B is the number of spectral bands, and T is the number of time nodes. The tth $(t = 1, 2, \ldots, T)$ time node of the image $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$ is denoted as $\mathcal{X}^t \in \mathbb{R}^{H \times W \times B}$. Similarly, the bth $(b = 1, 2, \ldots, B)$ spectral band and the tth $(t = 1, 2, \ldots, T)$ time node of the image $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$ is denoted as $\mathbf{X}_b^t \in \mathbb{R}^{H \times W}$.

B. Data-Aware Fidelity and Regularization

1) Data-Aware Fidelity: As the deep neural network implies multiscale feature maps in the process of encoding and decoding, it provides a basis for fully mining the deep semantics of RS images. The data-aware fidelity operates under the untrained deep domain, which captures the unique features of the multitemporal RS image by encouraging the matching of the multiscale deep semantics of the reconstructed image $\mathcal{X}$ and the guidance image $\mathcal{R}$. Since multitemporal RS images record the same geographical scene at different time nodes, similar information from various time nodes can be effectively utilized to assist in reconstruction. The guidance image $\mathcal{R}$ can be obtained by estimating the cloud-contaminated regions through cloud-free regions in the other temporal images.

First, given the observed image $\mathcal{Y} \in \mathbb{R}^{H \times W \times B \times T}$, suppose the time nodes $\{i_n\}_{n=1}^p$ are cloud-contaminated, while time

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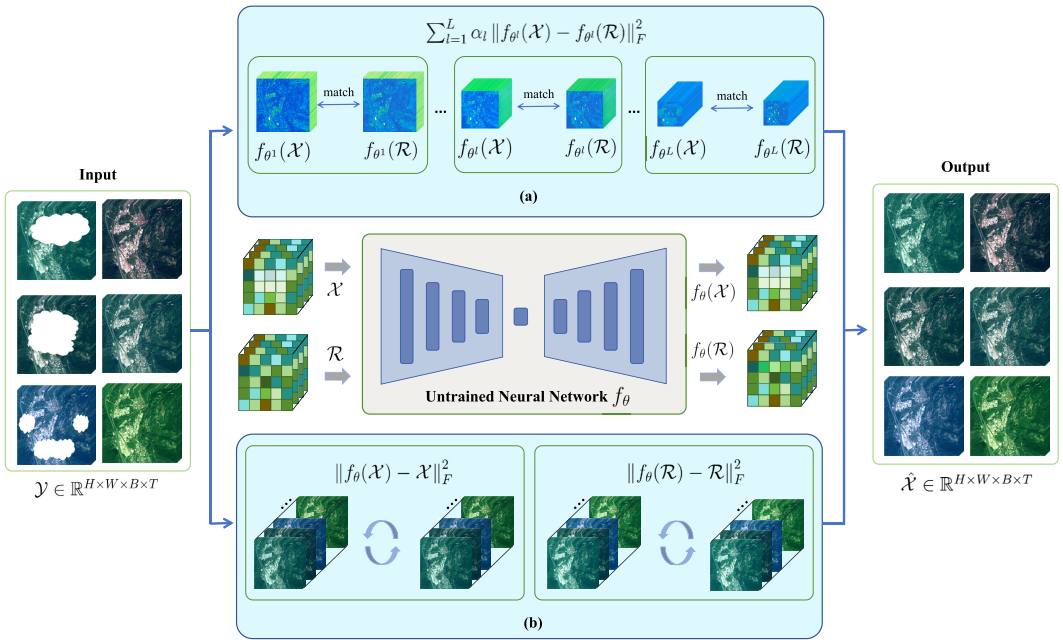


Fig. 2. Workflow of the proposed DAFRL. Leveraging an untrained neural network $f_\theta(\cdot)$ (e.g., U-Net), the unified DAFRL paradigm integrates two aspects into a single framework for cloud removal. $f_{\theta^l}(\cdot)$ is the feature under the first l layers ($l = 1, 2, \ldots, L$) of the untrained neural network $f_\theta(\cdot)$. (a) Data-aware fidelity. (b) Deep self-representation regularization.

nodes $\{i_m\}_{m=1}^{q}$ are cloud-free, where $p + q = T$. The average image $\bar{\mathcal{Y}} \in \mathbb{R}^{H\times W\times B}$ is computed by averaging the cloud-free images

$$\bar{\mathcal{Y}} = \frac{1}{q}\sum_{m=1}^{q} \mathcal{Y}^{j_m} \tag{1}$$

where $\mathcal{Y}^{j_m} \in \mathbb{R}^{H\times W\times B}$ represents the cloud-free image at j_mth time node.

Next, the relationship between cloud-contaminated images $\mathcal{Y}^{i_n} \in \mathbb{R}^{H\times W\times B}$ and average image $\bar{\mathcal{Y}} \in \mathbb{R}^{H\times W\times B}$ can be modeled by the following least-squares problem:

$$\min_{\mathbf{W}_b^t, c_b^t} \sum_{t=i_1}^{i_p}\sum_{b=1}^{B} \|\mathcal{P}_\Omega\left(\mathbf{Y}_b^t\right) - \left(\mathbf{W}_b^t \mathcal{P}_\Omega\left(\bar{\mathbf{Y}}_b\right) + c_b^t\right)\|_F^2 \tag{2}$$

where $\mathbf{Y}_b^t \in \mathbb{R}^{H\times W}$ represents the tth time node and the bth spectral band of the observed image $\mathcal{Y}$, $\bar{\mathbf{Y}}_b \in \mathbb{R}^{H\times W}$ represents the bth spectral band of the average image $\bar{\mathcal{Y}}$, and $\mathbf{W}_b^t$ and c_b^t are the coefficients of the least-squares problem. The set Ω indicates cloud-free regions, the set Ω^C indicates cloud-contaminated regions, and the projection operator $\mathcal{P}_\Omega(\cdot)$ retains the pixels at Ω while setting others to 0.

Building on this, the contaminated pixels in the image $\hat{\mathcal{Y}} \in \mathbb{R}^{H\times W\times B\times T}$ are estimated for each spectral band ($b = 1, 2, \ldots, B$) using the average image $\bar{\mathcal{Y}}$

$$\mathcal{P}_{\Omega^C}\left(\hat{\mathbf{Y}}_b^t\right) = \mathbf{W}_b^t \mathcal{P}_{\Omega^C}\left(\bar{\mathbf{Y}}_b\right) + c_b^t \text{ for } t = i_1, i_2, \ldots, i_p \tag{3}$$

where $\hat{\mathbf{Y}}_b^t \in \mathbb{R}^{H\times W}$ represents the estimated image $\hat{\mathcal{Y}}$ at the tth time node and bth spectral band.

Finally, the guidance image $\mathcal{R} \in \mathbb{R}^{H\times W\times B\times T}$ is constructed by replacing the cloud-contaminated regions in the observed multitemporal RS image $\mathcal{Y}$ with the corresponding regions from the estimated image $\hat{\mathcal{Y}}$

$$\mathcal{R} = \mathcal{P}_\Omega(\mathcal{Y}) + \mathcal{P}_{\Omega^C}\left(\hat{\mathcal{Y}}\right). \tag{4}$$

The data-aware fidelity captures features of an image through an end-to-end untrained neural network $f_\theta(\cdot)$ (e.g., U-Net). The feature map of the first l layers ($l = 1, 2, \ldots, L$) of the reconstructed image $\mathcal{X}$ is expressed through $f_{\theta^l}(\mathcal{X})$. Combined with the guidance image $\mathcal{R}$ estimated from observed multitemporal RS image, the data-aware fidelity can be expressed as follows:

$$\min_{\mathcal{X},\theta} \sum_{l=1}^{L} \alpha_l \|f_{\theta^l}(\mathcal{X}) - f_{\theta^l}(\mathcal{R})\|_F^2 \tag{5}$$

where $\mathcal{R}$ is the guidance image, $f_{\theta^l}(\mathcal{X})$ is the feature under the first l layers of untrained neural network, and $\alpha_l (l = 1, 2, \ldots, L)$ are the tradeoff parameters of the loss for the first l layers.

2) Deep Self-Representation Regularization: In the meantime, we suggest a deep self-representation based on the same untrained neural network $f_\theta(\cdot)$, which effectively captures the implicit regularization inherent in both the multitemporal RS images and the structural prior encoded within the network itself.

Self-representation models the relationship among all data samples by representing each point as a linear combination of other points within the same subspace, which is well-suited as a regularization term for addressing data corruption [45]. Empowered by deep neural networks, deep self-representation is able to mine the internal relationships of complex data to learn meaningful representations.

By jointly modeling the deep self-representation of both the reconstructed image $\mathcal{X}$ and the guidance image $\mathcal{R}$, the implicit regularization is formulated as

$$\min_{\mathcal{X},\theta} \beta_1 \|f_\theta(\mathcal{X}) - \mathcal{X}\|_F^2 + \beta_2 \|f_\theta(\mathcal{R}) - \mathcal{R}\|_F^2 \tag{6}$$

where $f_\theta(\mathcal{X})$ is the final output obtained by taking $\mathcal{X}$ as input through the complete untrained neural network, and β_1 and

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β_2 are the tradeoff parameters of the two parts of the implicit regularization.

The proposed deep self-representation generalizes the traditional self-representation model. Notably, when the untrained neural network is reduced to a single-layer linear structure without activation functions, the deep self-representation degenerates into the traditional self-representation model, as discussed in Remark 1.

Remark 1: Let a deep self-representation be defined as a composition of mappings

$$\begin{aligned}\mathcal{X} &= f_\theta(\mathcal{X}) \\ &= \psi^{-1}(\sigma_L(\mathcal{W}_L \Delta \sigma_{L-1}(\mathcal{W}_{L-1}\Delta \cdots \sigma_1(\mathcal{W}_1 \Delta \psi(\mathcal{X}))\cdots))) \end{aligned} \tag{7}$$

where $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$, $\psi(\cdot) : \mathbb{R}^{H\times W\times B\times T} \to \mathbb{R}^{H\times W\times \mathrm{BT}}$, $\psi^{-1}(\cdot)$ *denotes the inverse operation of* $\psi(\cdot)$, $\{\mathcal{W}_\ell \in \mathbb{R}^{r_\ell \times r_{\ell-1}\times \mathrm{BT}}\}_{\ell=1}^{L}$ *is a sequence of third-order tensors with* $r_0 = H, r_L = H$, Δ *denotes the face-wise product between two third-order tensors, and* $\sigma_\ell(\cdot)$ *are nonlinear activation functions applied on each frontal slice.*

When $L = 1$ *and* $\sigma_1(\cdot)$ *is set to the identity function, the proposed deep self-representation [see (7)] degenerates into the traditional self-representation*

$$\mathcal{X} = f_\theta(\mathcal{X}) = \psi^{-1}(\mathcal{W}_1 \Delta \psi(\mathcal{X})). \tag{8}$$

C. Proposed DAFRL Model and Solving Algorithm

The thick cloud removal task aims to reconstruct a cloud-free multitemporal RS image $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$ from a cloud-contaminated observation $\mathcal{Y} \in \mathbb{R}^{H\times W\times B\times T}$. Building upon the proposed fidelity and regularization components within an untrained neural network, we establish the unified DAFRL paradigm for thick cloud removal in multitemporal RS images

$$\begin{aligned}\min_{\mathcal{X},\theta}\ & \sum_{l=1}^{L}\frac{\alpha_l}{2}\|f_{\theta^l}(\mathcal{X}) - f_{\theta^l}(\mathcal{R})\|_F^2 \\ &+ \frac{\beta_1}{2}\|f_\theta(\mathcal{X}) - \mathcal{X}\|_F^2 + \frac{\beta_2}{2}\|f_\theta(\mathcal{R}) - \mathcal{R}\|_F^2 \\ \text{s.t.}\ & \mathcal{P}_\Omega(\mathcal{X}) = \mathcal{P}_\Omega(\mathcal{Y}) \end{aligned} \tag{9}$$

where $f_\theta(\cdot)$ is the untrained neural network (e.g., U-Net), $\mathcal{X}$ is the parameterized random input that is updated along with the network parameters, $\mathcal{R}$ is the guidance image, $f_{\theta^l}(\mathcal{X})$ is the feature under the first l layers ($l = 1, 2, \ldots, L$) of untrained neural network, $f_\theta(\mathcal{X})$ is the final output obtained by taking $\mathcal{X}$ as input through the complete untrained neural network, $\alpha_l (l = 1, 2, \ldots, L)$ are the tradeoff parameters of the loss for the untrained deep domain feature fidelity, and β_1 and β_2 are the tradeoff parameters of implicit regularization.

For solving the highly nonconvex and nonlinear problem, we adopt the adaptive moment estimation (Adam) optimizer to address the cloud removal model by optimizing the untrained neural network parameters θ and the output $\mathcal{X}$, and the problem (9) can be formulated as

$$\begin{aligned}\min_{\mathcal{X},\theta}\ & \sum_{l=1}^{L}\frac{\alpha_l}{2}\|f_{\theta^l}(\mathcal{X}) - f_{\theta^l}(\mathcal{R})\|_F^2 + \frac{\beta_1}{2}\|f_\theta(\mathcal{X}) - \mathcal{X}\|_F^2 \\ &+ \frac{\beta_2}{2}\|f_\theta(\mathcal{R}) - \mathcal{R}\|_F^2 + \frac{1}{2}\|\Psi(\mathcal{X} - \mathcal{Y})\|_F^2 \end{aligned} \tag{10}$$

where

$$\Psi(\mathcal{X} - \mathcal{Y}) := \begin{cases} 0, & \text{if } \mathcal{P}_\Omega(\mathcal{X}) = \mathcal{P}_\Omega(\mathcal{Y}), \\ \mathcal{X} - \mathcal{Y}, & \text{otherwise.} \end{cases} \tag{11}$$

To better illustrate the proposed method, we summarize the steps of our method for thick cloud removal in Algorithm 1.

Algorithm 1 Proposed DAFRL Model for Thick Cloud Removal

Input: The observed multi-temporal RS image $\mathcal{Y}$, the tradeoff parameters $\alpha = 1$, $\beta_1 = 10$, and $\beta_2 = 0.1$, and the indicator set Ω.

Initialization: $\mathcal{X}^{(0)}, \mathcal{Z}_1, \mathcal{Z}_2, k = 0$, and $k_{\max} = 10000$.

```
Compute Ȳ via (1).
for b = 1 : B do
    for t = i_1, i_2, ··· , i_p do
        Estimate W_b^t and c_b^t via (2).
        Estimate Ŷ via (3).
        Compute guidance image R via (4).
    end for
end for
while not converged and k < k_max do
    Updating θ^(k+1) and X^(k+1) in (10) using the Adam algorithm.
    Let k = k + 1.
end while
Compute X̂ = P_Ω(Y) + P_Ω^c(X^(k_max)).
```

Output: The reconstructed image $\hat{\mathcal{X}}$.

IV. Numerical Experiment

This section evaluates the performance of the proposed DAFRL for thick cloud removal in multitemporal RS images on both simulated and real data. To verify the superiority of the proposed method, we compare it with six relative methods, including regression, HaLRTC [46], TNN [47], TVTR [21], DIP [22], TRGFid [23], and DFTR [24]. Regression utilizes the information from cloud-free time nodes to reconstruct the image using the least squares method. HaLRTC and TNN methods reconstruct an image by using different hand-crafted low-rank regularization with pixel domain fidelity. TVTR method combines TV regularization with tensor ring decomposition. DIP method mines the prior information based on a deep CNN. Moreover, TRGFid and DFTR methods leverage different transform domain fidelity to remove the thick cloud. For the compared methods, the parameters are manually fine-tuned to achieve optimal performance according to the suggestions of the original article. Our experiments are conducted through MATLAB R2023a based on an Intel Core i7-12700 CPU with 32-GB RAM and the PyTorch 2.1.0 based on NVIDIA RTX 4060Ti GPU with 16-GB RAM.

A. Simulated Experiment

We implement simulated experiments to evaluate the performance of the proposed DAFRL using multitemporal RS

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TABLE I

QUANTITATIVE EVALUATION RESULTS BASED ON DIFFERENT METHODS UNDER DIFFERENT TIME NODES. THE **BEST** RESULTS ARE IN BOLD AND THE SECOND-BEST RESULTS ARE UNDERLINED

Region	Contaminated nodes	Metric	Observed	Regression	HaLRTC	TNN	TVTR	DIP	TRGFid	DFTR	DAFRL
France	Node 1	PSNR	6.698	47.144	42.987	45.215	39.855	44.676	47.279	47.335	**48.741**
		SSIM	0.7545	0.9933	0.9816	0.9903	0.9672	0.9824	0.9933	0.9935	**0.9950**
		SAM	4.2152	1.2115	1.0407	0.7316	1.4633	0.8824	0.5831	0.5025	**0.4528**
	Node 2	PSNR	5.868	45.505	40.915	44.155	37.925	43.142	45.790	45.902	**47.893**
		SSIM	0.7038	0.9904	0.9719	0.9867	0.9510	0.9760	0.9906	0.9907	**0.9941**
		SAM	4.8203	1.4005	1.4088	0.9503	1.9919	1.0309	0.6990	0.6728	**0.5431**
	Node 3	PSNR	7.747	46.313	47.360	43.626	44.494	48.107	47.528	48.730	**49.326**
		SSIM	0.7940	0.9951	0.9920	0.9938	0.9955	0.9822	0.9907	0.9960	**0.9970**
		SAM	3.2418	0.6707	0.5996	0.4499	0.9236	0.5067	0.3953	0.3242	**0.2735**
Morocco	Node 1	PSNR	12.394	46.175	46.217	48.015	42.663	51.226	51.817	52.401	**53.362**
		SSIM	0.8917	0.9980	0.9911	0.9961	0.9799	0.9968	0.9983	0.9982	**0.9983**
		SAM	2.5711	0.3109	0.1121	0.1196	0.1371	0.0872	0.0831	0.0760	**0.0737**
	Node 2	PSNR	7.3718	43.063	36.467	40.489	35.128	44.310	43.098	43.079	**44.626**
		SSIM	0.7096	0.9847	0.9400	0.9750	0.9087	0.9870	0.9846	0.9844	**0.9900**
		SAM	8.2847	0.3435	0.5465	0.3637	0.6327	0.3316	0.3343	0.2642	**0.2630**
	Node 3	PSNR	12.322	34.051	48.987	49.764	46.055	49.095	47.938	51.161	**52.844**
		SSIM	0.9125	0.9919	0.9933	0.9961	0.9856	0.9963	0.9964	0.9969	**0.9979**
		SAM	2.3310	0.5566	0.1051	0.1512	0.1077	0.0830	0.0706	0.0697	**0.0659**
Brazil	Node 1	PSNR	9.960	42.126	39.297	40.465	39.037	44.945	43.120	44.770	**46.576**
		SSIM	0.8295	0.9925	0.9693	0.9839	0.9567	0.9933	0.9856	0.9929	**0.9948**
		SAM	6.1414	0.7395	0.6171	0.4976	0.6161	0.2768	0.3478	0.2950	**0.2541**
	Node 2	PSNR	7.504	37.195	35.282	36.536	34.685	38.147	37.700	37.412	**40.407**
		SSIM	0.7301	0.9844	0.9433	0.9709	0.9263	0.9854	0.9707	0.9838	**0.9864**
		SAM	10.5420	0.7663	1.1272	0.9053	1.0040	0.4842	0.5410	0.3923	**0.3702**
	Node 3	PSNR	8.963	39.895	37.624	38.508	38.244	44.009	42.771	43.560	**45.504**
		SSIM	0.7958	0.9903	0.9592	0.9774	0.9473	0.9932	0.9927	0.9918	**0.9942**
		SAM	7.7506	0.9307	0.8457	0.7373	0.7768	0.3066	0.4274	0.3398	**0.2928**
	Average	PSNR	8.759	42.385	41.682	42.975	39.787	45.295	45.227	46.039	**47.698**
		SSIM	0.7913	0.9912	0.9735	0.9856	0.9576	0.9881	0.9892	0.9920	**0.9942**
		SAM	5.5442	0.7700	0.7114	0.5452	0.8504	0.4433	0.3868	0.3263	**0.2877**

images from Sentinel-2.[1] These datasets are available from the Copernicus Data Space Ecosystem. To quantitatively evaluate the performance of different methods, we employ three quantitative measures as evaluation indices, including the peak signal-to-noise ratio (PSNR), the structure similarity index measure (SSIM), and the spectral angle mapper (SAM). Typically, the higher PSNR and SSIM values, and the smaller SAM values indicate better cloud removal performance. More details and information related to the simulated experiments are provided below.

1) Dataset: To demonstrate the effectiveness and robustness of the proposed DAFRL, we select three datasets, which are taken from different places, including France, Brazil, and Morocco. Specifically, the datasets of *France* are a fourth-order tensor of size $600 \times 600 \times 4 \times 6$ with the spatial resolution of 10 m. Table II shows the details of the datasets used for the simulated experiment. The datasets of *Morocco* and *Brazil* are fourth-order tensors of size $400 \times 400 \times 4 \times 6$ with the spatial resolution of 10 m. The various datasets belong to different continents and have different landscape characteristics, including urban areas, rivers, hilly terrain, mountainous regions, etc. (as shown in Fig. 3).

[1]https://earthexplorer.usgs.gov

TABLE II

COMPREHENSIVE DETAILS ON MULTITEMPORAL RS IMAGES FOR THE THREE SIMULATED DATASETS

Dataset	Source	Image size	Spectral bands	Time nodes	Spatial resolution
France	Sentinel-2	600×600	4	6	10 m
Morocco	Sentinel-2	400×400	4	6	10 m
Brazil	Sentinel-2	400×400	4	6	10 m

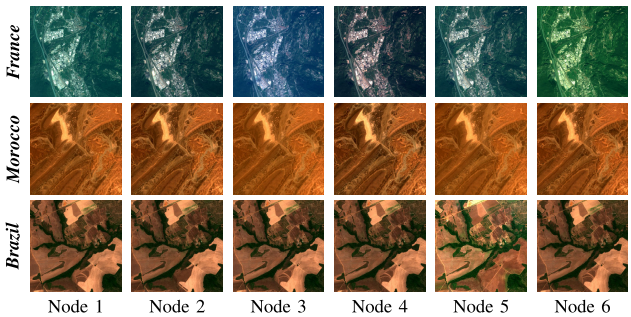


Fig. 3. Visual display (R: B3, G: B2, and B: B1) of multitemporal RS images for the simulated datasets.

We add simulated cloud masks at the first three time nodes of each dataset as contaminated nodes. The simulated cloud masks are set at different spatial locations across contaminated

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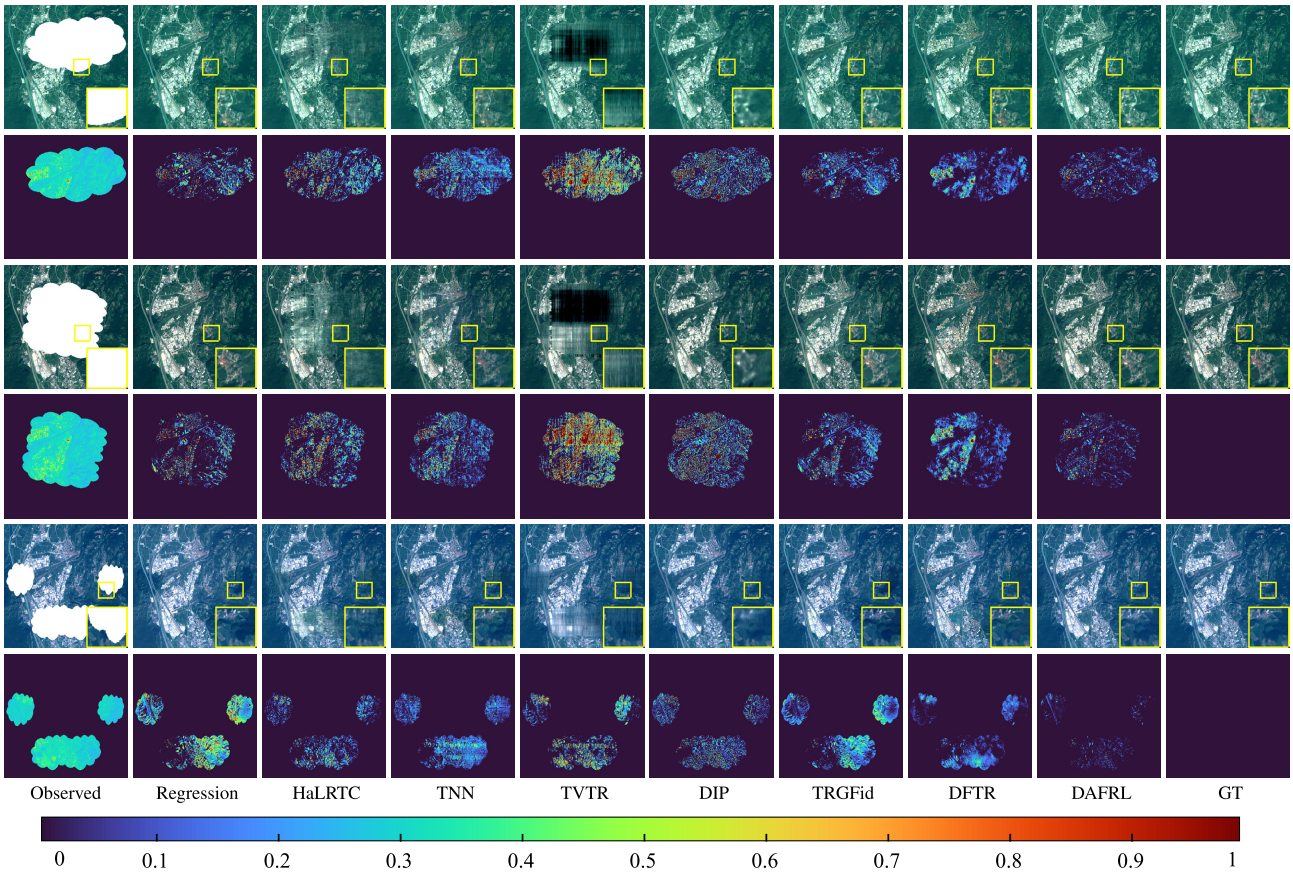


Fig. 4. Experimental results from various cloud removal methods of *France* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, and 3. Odd rows are images composed of bands (R:3, G:2, and B:1), and even rows are residual images composed of all bands averaged.

nodes and vary in size and shape. For a given contaminated node, the simulated cloud mask covers the same spatial locations across all spectral bands. To ensure a fair comparison, the same experimental design is applied to both the proposed method and the comparison methods.

2) Quantitative Comparison: The quantitative comparative results obtained by various cloud removal methods are shown in Table I. It can be observed that the proposed method achieves the best results on all datasets, which reveals the superiority of our DAFRL over the traditional cloud removal methods. The promising results of our method can be attributed to the power of our meticulously designed data-adaptive deep representation. TRGFid and DFTR show their competitiveness in the comparison methods is due to the careful design of the fidelity and regularization terms, but still suffer from the mismatch of the respective domains of these two terms. It is worth noting that the proposed DAFRL achieves an average PSNR that surpasses the second-best method by approximately 1.5 dB. This improvement is benefiting from the complementary nature of the fidelity and regularization terms, which work synergistically within the unified learning paradigm.

3) Qualitative Comparison: The qualitative results of cloud removal by different methods and ground truth (GT) are shown in Figs. 4–7. The regions marked by yellow rectangles in each true-color image (rendered using the red, green, and blue channels) are magnified, with the corresponding zoomed-in view positioned in the bottom-right corner to enhance the visual comparisons. Simultaneously, the residual image, which highlights the differences between each reconstructed image and the GT, is displayed below its corresponding image. It can be seen that the proposed DAFRL shows excellent reconstruction results for multitemporal RS images. Overall, the compared methods, including DIP, TRGFid, and DFTR, achieved notable reconstruction results. The DIP benefits from the powerful representational capability of deep CNNs to achieve good overall reconstruction performance, but there are still many flaws in the details. TRGFid and DFTR leverage more subtle fidelity terms to produce sharper textures, ranking just below the proposed DAFRL. In comparison, DAFRL not only leverages the structural properties of deep networks as a regularization term to ensure the preservation of overall structural integrity during reconstruction, but also exploits multiscale fidelity terms in the dynamic deep domain to reduce deviations in fine details.

To further evaluate the quality of the reconstruction, Fig. 8 presents scatter plots illustrating the relationship between the pixel values of reconstruction and GT within cloud-contaminated regions for each method. The closer the overall distribution of points is to the diagonal line, the greater the similarity between the reconstructed pixel values and the original pixel values. As observed in Fig. 8, the scatter plots of the images reconstructed by the proposed method exhibit a point distribution that aligns most closely with the diagonal line, providing an intuitive demonstration of the superiority of DAFRL. This observation is consistent with the visual results presented in Figs. 4–7 and the quantitative comparison summarized in Table I. The proposed DAFRL surpasses the

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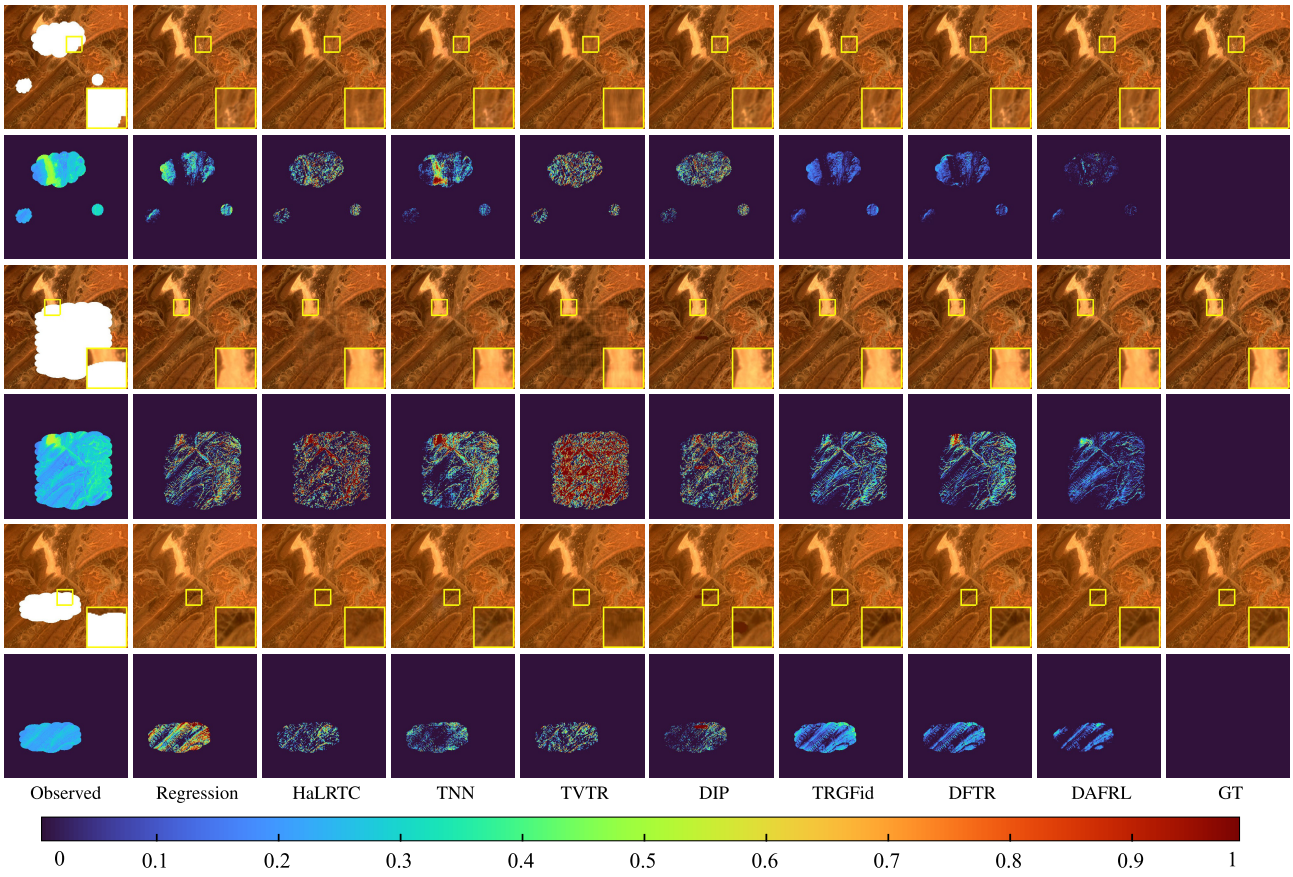


Fig. 5. Experimental results from various cloud removal methods of *Morocco* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, and 3. Odd rows are images composed of bands (R:3, G:2, and B:1), and even rows are residual images composed of all bands averaged.

TABLE III
COMPREHENSIVE DETAILS ON MULTITEMPORAL RS IMAGES FOR THE TWO REAL DATASETS

Dataset	Source	Image size	Spectral bands	Time nodes	Spatial resolution
Mantoval	Sentinel-2	600×600	4	4	10 m
Bourgogne	Landsat 8	600×600	7	4	30 m

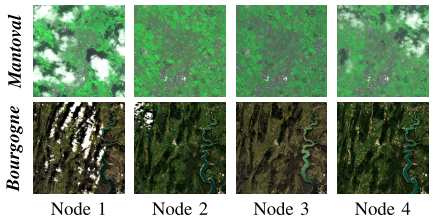


Fig. 6. Visual display of multitemporal RS images for the real datasets.

comparison methods by unifying the data-aware fidelity and implicit regularization terms within the learning paradigm. These characteristics enable DAFRL to deliver superior reconstruction results, balancing both global structure and detailed accuracy more effectively than other methods.

B. Real Experiments

To enable a more comprehensive comparison, we conduct real experiments to evaluate the performance of the proposed method. Two datasets from different satellites are selected, including *Mantoval* and *Bourgogne*. The details of these datasets are provided in Table III and the corresponding images are shown in Fig. 6. The *Mantoval* images, captured by Sentinel-2 over Mantoval, show cloud contamination at the first and fourth time nodes, with the cloud masks accessible via the website.[2] The *Bourgogne* images, captured by Landsat 8 over Bourgogne, show cloud contamination at the first and second time nodes, with both the dataset and cloud masks accessible via the website.[3] The remaining clean images are used as reference images to assist in reconstructing the images.

To enable effective visual comparison, the reconstruction results of all cloud removal methods are presented in Figs. 9 and 10, where the regions marked by red rectangles in each reconstructed image are enlarged for better comparison of image details. It is easy to observe that the proposed DAFRL obtains satisfactory results in these real experiments. The HaLRTC and TVTR methods exhibit insufficient capability for reconstructing fine details. Although TNN and DIP successfully reconstruct the overall structural integrity of the scenes, their reconstructed textures remain unsatisfactory. TRGFid and DFTR demonstrate improved preservation of edges and textures, but their performance in retaining intricate spatial features is still limited. In contrast, the proposed DAFRL achieves superior reconstruction quality, attributable to its data-aware fidelity from the untrained neural network, which effectively captures intrinsic features.

[2]https://earthexplorer.usgs.gov
[3]https://theia.cnes.fr/atdistrib/rocket/#/home

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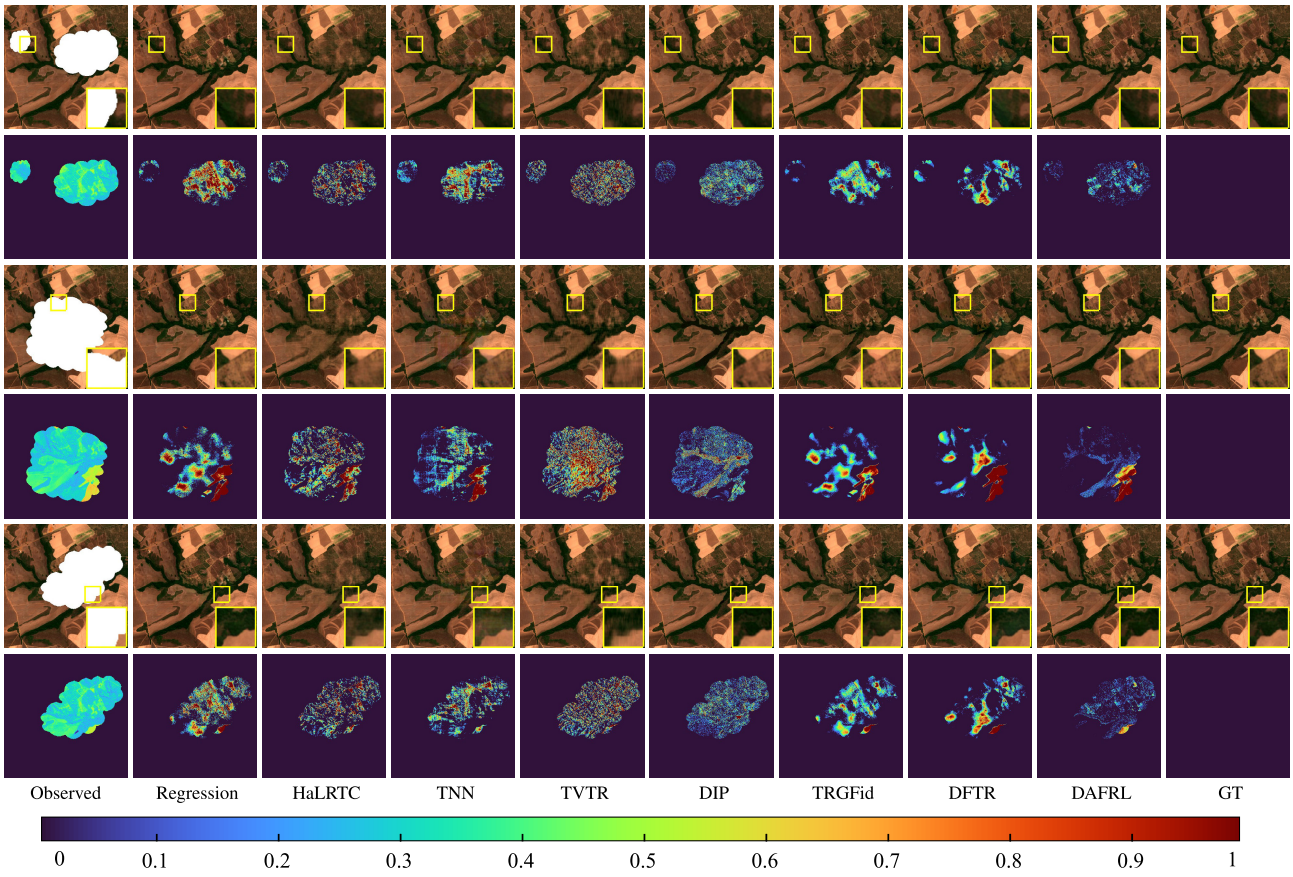


Fig. 7. Experimental results from various cloud removal methods of *Brazil* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, and 3. Odd rows are images composed of bands (R:3, G:2, and B:1), and even rows are residual images composed of all bands averaged.

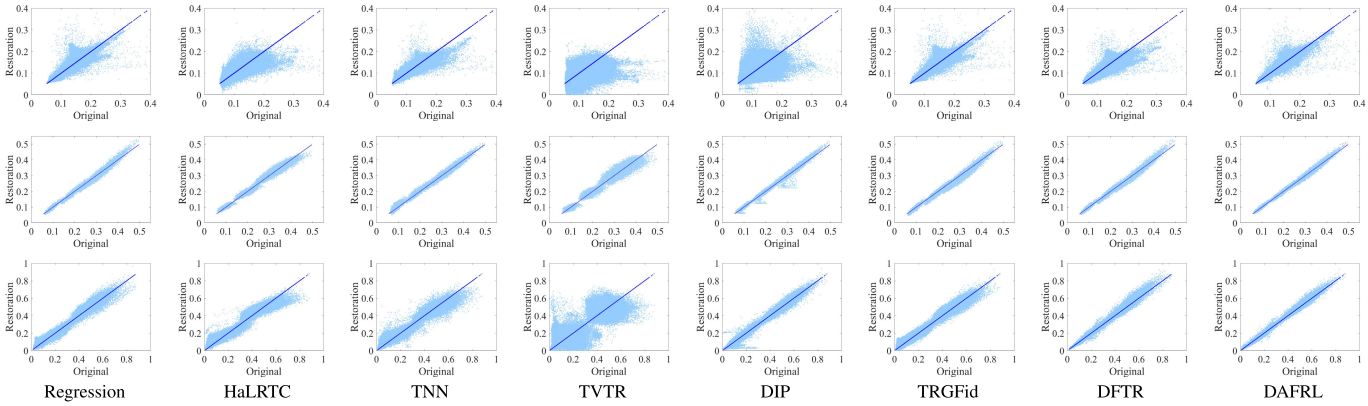


Fig. 8. Scatter plots comparing the original and reconstruction pixels in the missing areas for different simulated datasets. The datasets are shown from top to bottom as follows: *France* at contaminated node 2, *Morocco* at contaminated node 1, and *Brazil* at contaminated node 3.

C. Discussion

1) Contribution of Data-Aware Fidelity: In this section, we assess the effectiveness of the data-aware fidelity in our DAFRL. For this reason, we compare the data-aware fidelity with other existing fidelities in cloud removal. The reconstruction results obtained using different fidelities are presented in Table IV. It is observed that both fixed transform domain fidelity and data-aware fidelity perform better than the original pixel domain fidelity when reconstructing multitemporal RS images. In addition, the data-aware fidelity shows advantageous performances compared to the fixed transform fidelity. This improved performance can be attributed to the flexibility of untrained neural networks, which enable the data-aware fidelity to extract deep data-adaptive features. Approaches

TABLE IV

QUANTITATIVE COMPARISON FOR THE DIFFERENT FIDELITY ON FRANCE DATASET

Time node	Node 1			Node 2			Node 3		
Fidelity \ Metric	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Original Fidelity	44.876	0.9894	0.6038	44.733	0.9865	0.7605	47.644	0.9945	0.3840
Gradient Fidelity	47.139	0.9916	0.5545	45.778	0.9881	0.7521	47.678	0.9954	0.3542
Pretrained Fidelity	47.409	0.9935	0.4908	45.775	0.9907	0.6715	48.757	0.9959	0.3398
Data-Aware Fidelity	**48.741**	**0.9950**	**0.4528**	**47.893**	**0.9941**	**0.5431**	**49.326**	**0.9970**	**0.2735**

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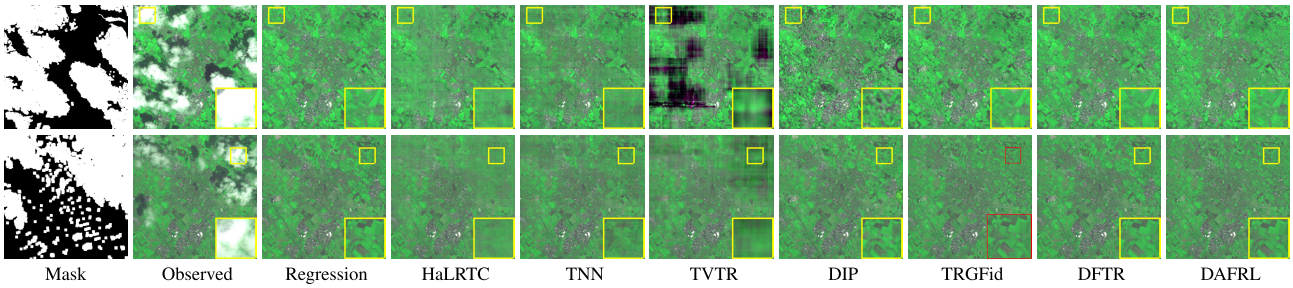


Fig. 9. Experimental results from various cloud removal methods of *Mantoval* dataset. The images are presented from top to bottom at contaminated nodes 1 and 2.

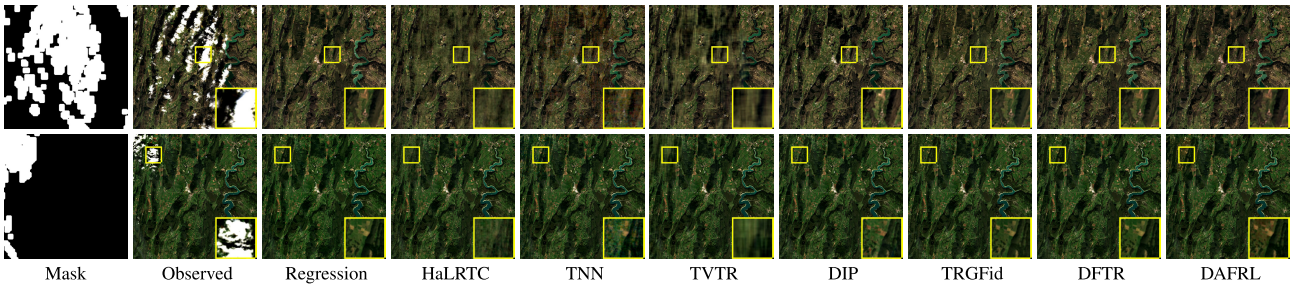


Fig. 10. Experimental results from various cloud removal methods of *Bourgogne* dataset. The images are presented from top to bottom at contaminated nodes 1 and 2.

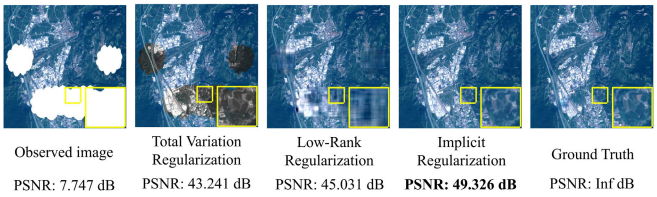


Fig. 11. Illustrating the effect of different regularization on *France* dataset.

TABLE V

QUANTITATIVE COMPARISON FOR THE DIFFERENT REGULARIZATION ON FRANCE DATASET

Time Node	Node 1			Node 2			Node 3		
Metric / Regularization	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
TV Regularization	41.148	0.9796	3.2794	42.252	0.9784	3.1888	43.241	0.9914	1.3823
Low-Rank Regularization	41.757	0.9732	1.1665	40.044	0.9608	1.5524	45.031	0.9860	0.7602
Implicit Regularization	**48.741**	**0.9950**	**0.4528**	**47.893**	**0.9941**	**0.5431**	**49.326**	**0.9970**	**0.2735**

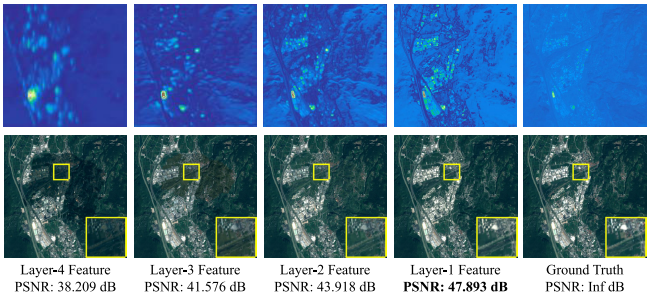


Fig. 12. Illustrating the influence of different layers of feature on *France* dataset. The first row is the feature maps at layer-4, layer-3, layer-2, and layer-1, respectively, which have been averaged along the band dimension. The second row is the corresponding reconstruction results.

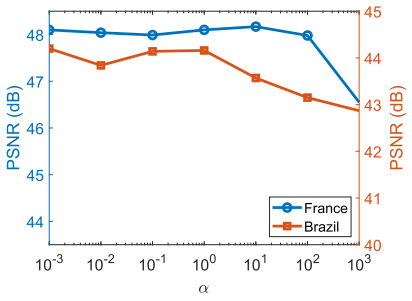


Fig. 13. Sensitivity analysis of the parameters of α on different datasets.

with classical fidelity can handle anomalous structures but may not be able to reconstruct fine details because they ignore the unique feature of each RS image. In contrast, the proposed data-aware fidelity improves the overall quality of the reconstructed images with better detail preservation.

2) Role of Implicit Regularization: To evaluate the role of implicit regularization, we compare the performance of different regularizations for cloud removal under the proposed model, and the results can be seen in the Table V and Fig. 11. Based on the observation, it can be learned that the implicit regularization derived from the deep self-representation can help to reconstruct the multitemporal RS image better than other regularizations. These conventional hand-crafted regularizations often fail to capture the rich structure that many natural signals exhibit, and the unique characteristics of individual RS images due to their limited characterization capability. Whereas the deep self-representation leverages the untrained neural network structure to extract the inherent prior of each RS image, serving as implicit regularization, and hence DAFRL is expected to be more effective in modeling the underlying structure of the data to achieve better reconstruction performance.

Furthermore, it becomes evident that the paradigm suffers from limitations in its representational capacity, whether by employing a different fidelity or utilizing another regularization. The reason for this shortcoming is that changing these two components destroys the integrity of the proposed paradigm. The data-independent fidelity and the implicit reg-

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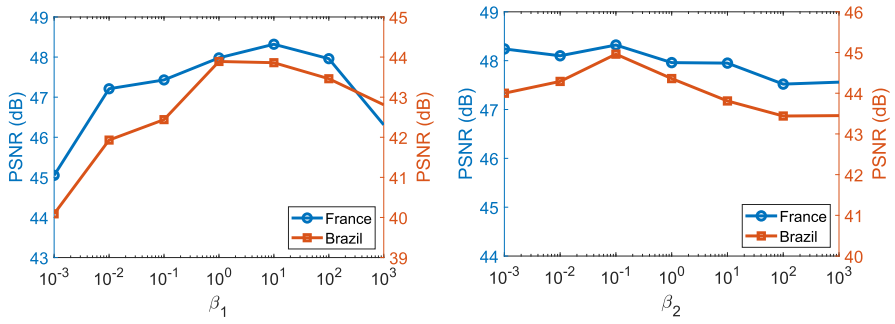


Fig. 14. Sensitivity analysis of the parameters of β_1 and β_2 on different datasets.

ularization are two sides of the same coin of the proposed paradigm, both rooted in the same untrained neural network. By collaborating and mutually reinforcing each other, the data-aware fidelity and implicit regularization enable cloud removal to outperform other combination methods. This synergy makes them an optimal pairing for high-quality reconstruction.

3) Analysis of the Features of Different Layers: This section analyses the impact of features extracted from different layers of the untrained neural network on the effectiveness of cloud removal. Taking a U-Net architecture with four encoder and four decoder layers as an example, the feature from the first encoder layer is referred to as the layer-1 feature, and so on, with the feature from the fourth encoder layer referred to as the layer-4 feature. Fig. 12 presents the feature maps from each layer along with their corresponding reconstruction results.

It can be observed that layer-1 features are most beneficial for cloud removal, whereas features from deeper layers offer limited contribution to the reconstruction of RS images. The features from deeper layers tend to capture semantic information and global context, which are typically more valuable for downstream tasks in general image processing. In contrast, the shallow features strike a better balance between semantic content and the preservation of essential structural details, making them particularly advantageous for reconstructing cloud-contaminated regions in RS images.

4) Influence of Reference Images: This section investigates the influence of the number of reference images on RS image reconstruction quality. Relying on a single reference image may work well for RS images captured close in time, but becomes less reliable when the temporal gap increases. Scene variations over time can cause discrepancies between the reference and cloud-contaminated images, reducing reconstruction accuracy. In contrast, using multiple reference images provides a more comprehensive temporal context, mitigating individual anomalies and improving overall reconstruction performance.

To evaluate this, we compare the performance of our DAFRL using one, two, and three reference images. Table VI presents experimental findings based on the *France* dataset, showing that additional reference images enhance the accuracy and quality of cloud removal. While a single reference image can provide reasonable reconstruction results, limited data constrains the accuracy of the proposed paradigm. Therefore, in our experiments, we maintain an equal number of reference and cloud-contaminated images to ensure a balanced and effective reconstruction process.

TABLE VI

QUANTITATIVE COMPARISON FOR THE EFFECT OF THE NUMBER OF REFERENCE IMAGES ON THE PROPOSED METHOD

Contaminated nodes	Metric	Observed	Number of reference images		
			one	two	three
Node 1	PSNR	6.698	45.319	46.245	**48.741**
	SSIM	0.7545	0.9858	0.9893	**0.9950**
	SAM	4.2152	0.7468	0.5882	**0.4528**
Node 2	PSNR	5.868	43.910	45.030	**47.893**
	SSIM	0.7038	0.9800	0.9858	**0.9941**
	SAM	4.8203	0.9913	0.7484	**0.5431**
Node 3	PSNR	7.747	47.427	48.790	**49.326**
	SSIM	0.7942	0.9917	0.9945	**0.9970**
	SAM	3.2418	0.5454	0.3862	**0.2735**
Average	PSNR	6.771	45.552	46.688	**48.653**
	SSIM	0.7508	0.9858	0.9899	**0.9954**
	SAM	4.0924	0.7612	0.5743	**0.4231**

5) Hyperparameter Analysis: This section analyzes the sensitivity of our method to hyperparameters. In the DAFRL framework, the key hyperparameters include the tradeoff parameter of data-aware fidelity (denoted as α) and the tradeoff parameters of implicit regularization (denoted as β_1 and β_2). To assess the impact of these parameters, we employed PSNR as the evaluation metric. Furthermore, we conducted experiments on both the *France* and *Brazil* datasets to ensure a comprehensive evaluation and assess the generalizability of our approach across different geographic regions.

First, we examine the effect of tradeoff parameter α, which regulates the balance between the data-aware fidelity term and the other terms in the objective function. The test range for α is selected from the candidate set {0.001, 0.01, 0.1, 1, 10, 100, 1000}. As shown in Fig. 13, the reconstruction performance exhibits slight fluctuations as α increases from 0.001 to 1, followed by a decline in effectiveness at higher values. Empirically, we set the hyperparameter α to 1 in the proposed DAFRL method.

Next, we analyze the impact of the tradeoff parameters β_1 and β_2, which determine the significance of the implicit regularization terms. The test ranges for β_1 and β_2 are selected from the candidate set {0.001, 0.01, 0.1, 1, 10, 100, 1000}. As shown in Fig. 14, the reconstruction performance initially improves and then declines as β_1 increases, indicating an optimal range for this parameter. In contrast, β_1 exhibits a relatively stable effect across different values. Based on these observations, we empirically set β_1 to 10 and β_2 to 0.1 to

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achieve a favorable balance between regularization strength and reconstruction quality.

V. Conclusion

In this article, we have proposed a novel thick cloud removal framework, DAFRL, which integrates data-aware fidelity and deep self-representation regularization within a unified learning paradigm. This design overcomes the limitations of conventional methods that rely on handcrafted or pretrained filters, leading to enhanced detail preservation and more consistent reconstruction across diverse multitemporal RS images. By leveraging the adaptability of untrained neural networks, our method enables deep data-aware feature matching between contaminated and guidance images while simultaneously imposing implicit regularization. Both fidelity and regularization originate from the same untrained neural network, serving distinct yet complementary functions: ensuring accurate feature alignment while enhancing structural coherence, thereby synergistically improving reconstruction quality. Extensive experiments on both simulated and real datasets demonstrate the superiority of DAFRL over existing methods, highlighting its effectiveness in maintaining fine RS image details.

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